

Guideline Title: Thoracic Aortic Dissection

Guideline Number: GUID-3203-M018

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
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Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** To recognize the patient with a thoracic aortic dissection and prevent further extension of the dissection. To minimize the sheer forces placed on the aortic wall.
- II. **DEFINITIONS:**
 - a. SBP – Systolic Blood Pressure
 - b. HR – Heart rate
- III. **DEPARTMENT GUIDELINE:**
 - a. Establish care as defined by GUID3203-G002 – *General Patient Care*
 - b. Obtain and record the Classification: Type A or Type B
 - c. Bring any available CT scan reports and images from sending facility.
 - d. Assess for distal pulses and neurological deficits. Obtain bilateral upper extremities BPs.
 - e. Treat pain per GUID3203-G005 *Analgesia and Sedation Management* and anxiety per GUID3203-G006 *Anxiety (Combative Patient Management)*.
 - f. Goals: adequate end-organ perfusion with SBP < 120 mmHg and HR < 60 bpm.
 - g. If SBP > 120 mmHg and HR > 60 bpm administer Beta Blockers
 - i. **Esmolol bolus 500 mcg/kg/min IV** over 1 minute followed by **Esmolol infusion at 50 mcg/kg/min**. Titrate every 5 minutes by repeating 500 mcg/kg/min bolus over 1 minute and then increasing infusion by 50 mcg/kg/min. Maximum dose 300 mcg/kg/min
 - ii. **Labetalol 10-20 mg IV** over 2 minutes. May repeat every 10 minutes until goal blood pressure obtained or maximum total dose of 300mg.
 - h. If SBP remains > 120 mmHg after β -blockers, or HR < 60 bpm prior to Beta Blockers administer:
 - i. **Nicardipine infusion at 5 mg/h**. Increase by 2.5 mg/h IV every 5-15 minutes to maximum of 15 mg/h.
 - ii. **Nitroprusside (50 mg/250ml D5W) infusion at 0.3 mcg/kg/min**. The infusion can be increased by 0.5 mcg/kg/min every 5 minutes until desired BP is reached. Max dose 10 mcg/kg/min (Avoid in renal failure).
 1. Treat reflex tachycardia from nitroprusside with labetalol or esmolol as above
 - i. If hypotension with signs of end-organ hypoperfusion (change in mental status, oliguria, etc)
 - i. Discontinue all anti-hypertensive medications.

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- ii. Administer IV fluid as per *GUID3203-M011 - Hypotension* and consider the use of blood products per *GUID3203-PR002 Blood Product Administration* or *GUID3203-PR025 Emergency Blood Product Administration*.
 - iii. Consider contacting managing physician for treatment guidelines for hypotension and use of vasopressors.
 - j. Patients with associated MI confirmed by ECG
 - i. Proceed as above with β -blockade.
 - ii. Avoid thrombolytics, heparin, and aspirin.
- IV. **DOCUMENTATION:**
- a. EMS Charts
- V. **REFERENCES:**